

Practical no:3

Aim: Comparison of Classification Accuracy of SVM for given dataset

Theory:

Support Vector Machines (SVMs) are a set of supervised learning methods used for classification, regression, and outliers detection. The primary goal of the SVM algorithm is to find a hyperplane in an N-dimensional space (N is the number of features) that distinctly classifies the data points.

Key concepts:

- **Hyperplane:** In SVM, a hyperplane is a decision boundary that helps classify the data points. Data points falling on either side of the hyperplane can be attributed to different classes.
- **Support Vectors:** Data points that are closer to the hyperplane and influence the position and orientation of the hyperplane. These are the critical elements of the dataset.
- **Margin:** The distance between the hyperplane and the closest data points from either class. SVM aims to maximize this margin.
- **Import Libraries:** Import necessary libraries such as numpy, pandas, matplotlib, and sklearn.
- **Load the Dataset:** Load and explore the dataset to understand its structure and features.
- **Preprocess the Data:** Handle missing values, encode categorical variables, and split the data into training and testing sets.
- **Train the SVM Model:** Train the SVM model using different kernel functions (e.g., linear, polynomial, RBF).
- **Evaluate the Model:** Evaluate the model's performance using classification accuracy and other relevant metrics.
- **Compare Results:** Compare the classification accuracy of the SVM model with different kernel functions.

Program:

```
#Comparison of Classification Accuracy of SVM for given dataset using  
iris dataset
```

```
import numpy as np  
from sklearn import datasets  
from sklearn.model_selection import train_test_split  
from sklearn.svm import SVC  
from sklearn.metrics import accuracy_score
```

```
# Load Iris dataset  
iris = datasets.load_iris()  
X = iris.data y =  
iris.target  
  
# Split the data into training and test sets  
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.5,  
random_state=62)  
  
# Define SVM models with different kernels kernels  
= ['linear', 'poly', 'rbf', 'sigmoid'] accuracies =  
{}  
  
for kernel in kernels:  
    # Initialize and train the SVM model  
    model = SVC(kernel=kernel, random_state=62)  
    model.fit(X_train, y_train)  
  
    # Make predictions  
    y_pred = model.predict(X_test)  
  
    # Calculate accuracy  
    accuracy = accuracy_score(y_test, y_pred)  
    accuracies[kernel] = accuracy  
    print(f'Accuracy with {kernel} kernel: {accuracy:.14f}')  
  
# Display results  
print("\nComparison of Classification Accuracy:") for  
kernel, accuracy in accuracies.items():  
    print(f"Kernel: {kernel} - Accuracy: {accuracy
```

output:

Accuracy with linear kernel: 0.9778

Accuracy with poly kernel: 0.9556

Accuracy with rbf kernel: 1.0000

Accuracy with sigmoid kernel: 0.8889

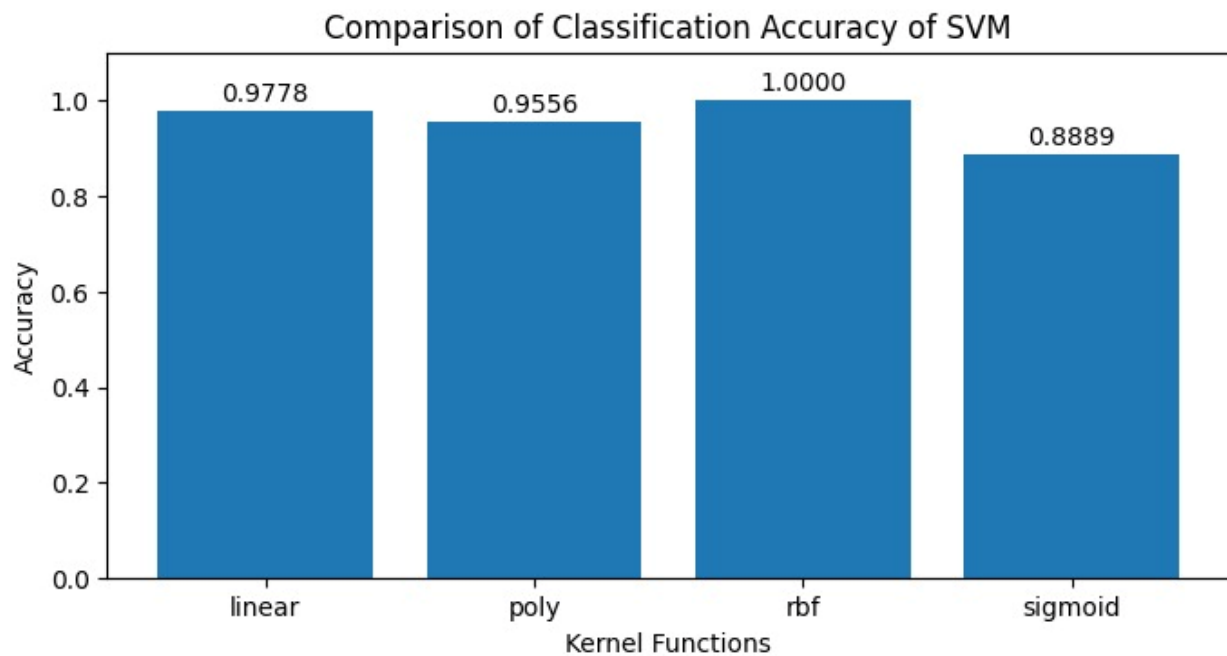
Comparison of Classification Accuracy:

Kernel: linear - Accuracy: 0.9778

Kernel: poly - Accuracy: 0.9556

Kernel: rbf - Accuracy: 1.0000

Kernel: sigmoid - Accuracy: 0.8889



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